

## Long Questions

### a. What is a Computer Network? Write any two advantages and limitations of it.

A **computer network** is a collection of two or more computers and other devices connected together through communication media to share data, resources, and services. Networks allow users to communicate and exchange information efficiently. Examples include school computer labs, office networks, and the Internet.

#### Advantages:

1. **Resource Sharing:** Devices such as printers, scanners, storage devices, and software can be shared among multiple users, reducing costs.
2. **Fast Communication:** Users can exchange information quickly through emails, messaging, video conferencing, and file sharing.

#### Limitations:

1. **Security Risks:** Networks are vulnerable to hacking, malware, viruses, and unauthorized access.
2. **High Setup and Maintenance Cost:** Large networks require networking devices, cables, software, and skilled personnel for maintenance.

### b. What are the Primary Purposes of Computer Networks?

The main purposes of computer networks are:

1. **Resource Sharing:** Allows users to share hardware, software, and data.
2. **Communication:** Enables communication through email, chat, video calls, and social media.

3. **Data Sharing:** Facilitates easy exchange of files and information among users.
4. **Centralized Management:** Administrators can manage resources and security from a central location.
5. **Internet Access:** Multiple users can access the Internet through a common network.
6. **Remote Access:** Users can access data and applications from distant locations.

Thus, computer networks improve efficiency, collaboration, and productivity.

### **c. Explain the Difference Between Simplex, Half-Duplex, and Full-Duplex Data Transmission Modes.**

#### **1. Simplex Mode**

In simplex communication, data flows in only one direction. One device only sends data while the other only receives it.

**Examples:** Keyboard to computer, television broadcasting. -

#### **2. Half-Duplex Mode**

In half-duplex communication, data can flow in both directions, but not at the same time. Devices take turns sending and receiving data.

**Examples:** Walkie-talkies.

#### **3. Full-Duplex Mode**

In full-duplex communication, data can flow in both directions simultaneously. Both devices can send and receive data at the same time.

**Examples:** Telephone calls, video conferencing.

| Feature                    | Simplex | Half-Duplex   | Full-Duplex |
|----------------------------|---------|---------------|-------------|
| Direction                  | One-way | Two-way       | Two-way     |
| Simultaneous Communication | No      | No            | Yes         |
| Speed                      | Lowest  | Medium        | Highest     |
| Example                    | TV      | Walkie-talkie | Telephone   |

#### **d. Differentiate Between Guided and Unguided Communication Channels.**

Communication channels are pathways through which data travels from sender to receiver.

##### **Guided Communication Channel**

A guided channel uses physical cables or wires to transmit data.

**Examples:** Twisted pair cable, coaxial cable, optical fiber.

##### **Characteristics:**

- Uses physical medium.
- More secure.
- Less affected by interference.
- Limited mobility.

##### **Unguided Communication Channel**

An unguided channel transmits data through air using electromagnetic waves.

**Examples:** Radio waves, microwaves, satellite communication.

##### **Characteristics:**

- No physical cable required.
- Supports mobility.
- Easier installation.
- More prone to interference.

| <b>Guided Media</b> | <b>Unguided Media</b>     |
|---------------------|---------------------------|
| Uses cables         | Uses air/wireless signals |
| More secure         | Less secure               |
| Limited mobility    | High mobility             |
| Lower interference  | Higher interference       |

#### **e. Explain the Use of Cat 6 Cable.**

**Category 6 (Cat 6) cable** is a standardized twisted-pair Ethernet cable used in computer networks for high-speed data transmission.

#### **Uses:**

1. Connecting computers to switches and routers.
2. Supporting Gigabit Ethernet networks.
3. Providing high-speed Internet connectivity.
4. Connecting network devices in offices, schools, and homes.
5. Supporting video streaming and large data transfers.

#### **Features:**

- Supports speeds up to **10 Gbps** over shorter distances.
- Reduces signal interference and crosstalk.
- Provides better performance than Cat 5 and Cat 5e cables.

Cat 6 cables are widely used in modern Local Area Networks (LANs).

## **f. Describe the Role of Satellites in Communication.**

Communication satellites are artificial satellites placed in orbit around the Earth to transmit and receive communication signals.

### **Roles of Satellites:**

1. **Television Broadcasting:** Deliver TV signals to different regions.
2. **Internet Services:** Provide Internet access in remote areas.
3. **Telephone Communication:** Facilitate long-distance and international calls.
4. **Weather Monitoring:** Collect weather data and assist forecasting.
5. **Navigation:** Support GPS and location-based services.
6. **Disaster Communication:** Maintain communication during emergencies.

Satellites help connect people across the world, especially where wired communication is difficult.

## **g. What is a Media Converter, and Why is it Used?**

A **media converter** is a networking device that converts one type of transmission media into another.

### **Uses:**

1. Converts electrical signals to optical signals and vice versa.
2. Connects fiber optic networks with copper cable networks.
3. Extends network distance beyond copper cable limitations.
4. Integrates different network technologies.
5. Improves network flexibility and scalability.

### **Example:**

A company may use a media converter to connect a copper Ethernet LAN to a fiber optic backbone.

### **h. Differentiate Between a Hub and a Switch.**

A hub and a switch are networking devices used to connect multiple computers.

#### **Hub**

A hub broadcasts incoming data to all connected devices regardless of the destination.

#### **Switch**

A switch sends data only to the intended destination device by using MAC addresses.

| <b>Feature</b>     | <b>Hub</b>                | <b>Switch</b>            |
|--------------------|---------------------------|--------------------------|
| Intelligence       | Less intelligent          | Intelligent              |
| Data Transmission  | Broadcasts to all devices | Sends to specific device |
| Speed              | Slower                    | Faster                   |
| Security           | Lower                     | Higher                   |
| Traffic Management | Poor                      | Efficient                |

Switches are preferred in modern networks due to better performance and security.

### **i. Explain the Function of a Repeater.**

A **repeater** is a network device used to regenerate and strengthen weak signals in a communication network.

**Functions:**

1. Receives weak or degraded signals.
2. Cleans and amplifies the signals.
3. Retransmits the signals to their destination.
4. Extends the communication distance between devices.
5. Reduces data loss caused by signal attenuation.

**Importance:**

- Helps maintain signal quality.
- Enables long-distance communication.
- Improves network coverage.

**Example:**

In a large office building, repeaters are installed to extend Ethernet or Wi-Fi signals to distant areas.

**j. Define Network Topology and Describe the Most Common Types Such as Bus, Star, and Ring.**

**Network topology** refers to the physical or logical arrangement of computers, cables, and networking devices in a network.

**Bus Topology**

All devices are connected to a single central cable called a backbone.

**Advantages:**

- Easy installation.
- Low cost.

**Disadvantages:**

- Backbone failure affects the entire network.

## **Star Topology**

All devices connect to a central hub or switch.

### **Advantages:**

- Easy management.
- Failure of one device does not affect others.

### **Disadvantages:**

- Central device failure affects the network.

## **Ring Topology**

Devices are connected in a circular manner.

### **Advantages:**

- Organized data flow.
- Equal access to resources.

### **Disadvantages:**

- Failure of one device can disrupt communication.

## **k. Discuss the Advantages and Limitations of Bus Topology.**

### **Advantages:**

1. Simple to install and configure.
2. Requires less cable than other topologies.
3. Cost-effective for small networks.
4. Easy to extend.

### **Limitations:**

1. Backbone cable failure stops the entire network.

2. Difficult troubleshooting.
3. Performance decreases with increased traffic.
4. Limited cable length.
5. Not suitable for large networks.

Bus topology is suitable for small networks but is rarely used today.

### **l. Compare Star and Hybrid Topologies.**

| <b>Feature</b>  | <b>Star Topology</b> | <b>Hybrid Topology</b>    |
|-----------------|----------------------|---------------------------|
| Structure       | Central hub/switch   | Combination of topologies |
| Complexity      | Simple               | Complex                   |
| Cost            | Moderate             | High                      |
| Scalability     | Good                 | Excellent                 |
| Fault Isolation | Easy                 | Easy                      |
| Flexibility     | Limited              | Very flexible             |

### **Star Topology**

All devices connect to one central device.

### **Hybrid Topology**

Combines two or more topologies such as star, bus, or ring.

Hybrid topology is suitable for large organizations requiring flexibility and scalability.

### **m. Explain the Difference Between LAN and WAN.**

#### **LAN (Local Area Network)**

A LAN connects devices within a limited area such as a home, school, or office.

## **WAN (Wide Area Network)**

A WAN connects devices across large geographical areas such as cities, countries, or continents.

| <b>Feature</b> | <b>LAN</b>     | <b>WAN</b>     |
|----------------|----------------|----------------|
| Coverage Area  | Small          | Large          |
| Speed          | Higher         | Lower          |
| Cost           | Lower          | Higher         |
| Ownership      | Private        | Public/Private |
| Example        | School network | Internet       |

### **n. Describe the Uses of MAN.**

A **Metropolitan Area Network (MAN)** connects multiple LANs within a city or metropolitan area.

#### **Uses:**

1. Connecting branch offices in a city.
2. Providing city-wide Internet services.
3. Interconnecting educational institutions.
4. Supporting government offices.
5. Connecting hospitals and healthcare centers.

MAN provides high-speed communication across urban areas.

### **o. Explain PAN with an Example.**

A **Personal Area Network (PAN)** is a network that connects personal devices within a short range, typically around 10 meters.

**Characteristics:**

- Small coverage area.
- Low cost.
- Often uses Bluetooth or USB connections.

**Example:**

A smartphone connected to wireless earbuds and a smartwatch through Bluetooth forms a PAN.

PAN is commonly used for personal communication and device synchronization.

**p. What is Network Architecture, and Why is it Important in Network Design?**

**Network architecture** is the overall design and structure of a computer network, including hardware, software, communication protocols, and transmission methods.

**Importance:**

1. Ensures efficient communication.
2. Improves network performance.
3. Enhances security.
4. Supports future expansion.
5. Simplifies management and maintenance.
6. Reduces operational costs.

A well-designed network architecture ensures reliability, scalability, and efficient resource utilization.

**q. Compare and Contrast the Client-Server and Peer-to-Peer Network Architectures.****Client-Server Architecture**

A central server provides services and resources to client computers.

**Advantages:**

- Better security.
- Centralized management.
- Easy backup.

**Disadvantages:**

- Expensive.
- Server failure affects clients.

**Peer-to-Peer Architecture**

All computers act as both clients and servers.

**Advantages:**

- Easy setup.
- Low cost.
- No dedicated server required.

**Disadvantages:**

- Less secure.
- Difficult management.

| Feature    | Client-Server    | Peer-to-Peer        |
|------------|------------------|---------------------|
| Server     | Dedicated server | No dedicated server |
| Security   | High             | Lower               |
| Cost       | Higher           | Lower               |
| Management | Centralized      | Decentralized       |

#### **r. Explain IPv4 with an Example.**

**IPv4 (Internet Protocol Version 4)** is the fourth version of the Internet Protocol used to identify devices on a network.

##### **Features:**

- Uses a 32-bit address.
- Consists of four decimal numbers separated by dots.
- Supports about 4.3 billion unique addresses.

##### **Example:**

**192.168.1.10**

In this address:

- 192.168.1 = Network portion
- 10 = Host portion

IPv4 enables devices to communicate over networks and the Internet.

#### **s. Describe IPv6 and Its Advantages.**

**IPv6 (Internet Protocol Version 6)** is the latest version of the Internet Protocol developed to overcome IPv4 address limitations.

##### **Example:**

**2001:0db8:85a3:0000:0000:8a2e:0370:7334**

##### **Advantages:**

1. Vast address space (128-bit addressing).
2. Improved security through built-in IPsec.
3. Better routing efficiency.
4. Automatic address configuration.

5. Reduced network congestion.
6. Improved support for mobile devices.

IPv6 ensures sufficient addresses for future Internet growth.

#### **t. What is a MAC Address?**

A **MAC (Media Access Control) address** is a unique hardware address assigned to a network interface card (NIC).

#### **Characteristics:**

- Usually 48 bits long.
- Assigned by manufacturers.
- Used for communication within local networks.

#### **Example:**

**00:1A:2B:3C:4D:5E**

#### **Functions:**

1. Identifies devices uniquely.
2. Helps switches forward data correctly.
3. Supports local network communication.

A MAC address acts like a permanent identification number for a network device.

#### **u. Define an Intranet and an Extranet, and Highlight Their Differences.**

##### **Intranet**

An intranet is a private network accessible only to authorized members of an organization.

## **Uses:**

- Internal communication.
- Sharing documents.
- Employee collaboration.

## **Extranet**

An extranet is an extension of an intranet that allows limited access to external users such as customers, suppliers, or partners.

## **Differences**

| <b>Feature</b> | <b>Intranet</b>        | <b>Extranet</b>            |
|----------------|------------------------|----------------------------|
| Users          | Employees only         | Employees and outsiders    |
| Access         | Internal               | Controlled external access |
| Security       | High                   | Very high                  |
| Purpose        | Internal communication | External collaboration     |

## **v. Write About Any Four Services of the Internet in Your Daily Life.**

### **1. Email**

Used for sending and receiving messages, documents, and files instantly.

### **2. Web Browsing**

Allows access to information, news, educational materials, and websites.

### **3. Online Banking**

Enables money transfers, bill payments, and account management from anywhere.

## 4. Video Conferencing

Supports online meetings, classes, and communication through platforms such as Zoom and Google Meet.

These services make communication, learning, and financial transactions more convenient.

### w. What is a Network Protocol, and Why is it Essential for Communication?

A **network protocol** is a set of rules and standards that govern how data is transmitted, received, and interpreted between devices on a network.

#### Importance of Network Protocols:

1. Ensure accurate data transmission.
2. Enable communication between different devices and operating systems.
3. Provide error detection and correction.
4. Enhance security and reliability.
5. Standardize communication processes.

#### Examples:

- **HTTP/HTTPS:** Web communication.
- **FTP:** File transfer.
- **SMTP:** Email transmission.
- **TCP/IP:** Internet communication.

Without network protocols, devices would not understand each other, making network communication impossible.

### 3. Write the technical term for the following:

| No. | Statement                                                             | Technical Term                      |
|-----|-----------------------------------------------------------------------|-------------------------------------|
| a   | Transmission of text, audio, and video through some electronic means. | <b>Telecommunication</b>            |
| b   | Data transmission is only in one direction.                           | <b>Simplex</b>                      |
| c   | Transmission of data both ways, but one at a time.                    | <b>Half-Duplex</b>                  |
| d   | Transmission of data in both directions simultaneously.               | <b>Full-Duplex</b>                  |
| e   | An interconnection between two or more computers.                     | <b>Computer Network</b>             |
| f   | A wire made of copper twisted together and insulated.                 | <b>Twisted Pair Cable</b>           |
| g   | Cable with a central conductor, insulating layer, and shield.         | <b>Coaxial Cable</b>                |
| h   | A cable that carries data as light waves.                             | <b>Fiber Optic Cable</b>            |
| i   | Radio waves for audio transmission.                                   | <b>Radio Communication</b>          |
| j   | High-frequency waves for long-distance communication.                 | <b>Microwaves</b>                   |
| k   | Provides a port to connect a computer to a network.                   | <b>Network Interface Card (NIC)</b> |
| l   | A device used to connect a PC to a telephone line.                    | <b>Modem</b>                        |
| m   | The computer on which users run applications.                         | <b>Client</b>                       |

| No. | Statement                                                          | Technical Term                        |
|-----|--------------------------------------------------------------------|---------------------------------------|
| n   | Each computer or device on a network.                              | <b>Node</b>                           |
| o   | A device that controls two dissimilar networks.                    | <b>Gateway</b>                        |
| p   | A computer that provides services to other computers.              | <b>Server</b>                         |
| q   | An operating system that can handle a network.                     | <b>Network Operating System (NOS)</b> |
| r   | An important software that controls and manages computer networks. | <b>Network Operating System (NOS)</b> |
| s   | Physical structure of a computer network.                          | <b>Network Layout</b>                 |
| t   | An interconnected pattern of network components.                   | <b>Network Topology</b>               |
| u   | The smallest type of computer network.                             | <b>PAN (Personal Area Network)</b>    |
| v   | Largest computer network.                                          | <b>WAN (Wide Area Network)</b>        |
| w   | Privately owned network of computers.                              | <b>Intranet</b>                       |
| x   | Network of networks.                                               | <b>Internet</b>                       |
| y   | Sending and receiving messages electronically via the Internet.    | <b>E-mail (Electronic Mail)</b>       |

#### 4. Write the full form of:

| No. | Abbreviation | Full Form                     |
|-----|--------------|-------------------------------|
| a   | LAN          | Local Area Network            |
| b   | WAN          | Wide Area Network             |
| c   | MAN          | Metropolitan Area Network     |
| d   | PAN          | Personal Area Network         |
| e   | UTP          | Unshielded Twisted Pair       |
| f   | STP          | Shielded Twisted Pair         |
| g   | BNC          | Bayonet Neill–Concelman       |
| h   | RJ-45        | Registered Jack-45            |
| i   | NIC          | Network Interface Card        |
| j   | ISP          | Internet Service Provider     |
| k   | WWW          | World Wide Web                |
| l   | URL          | Uniform Resource Locator      |
| m   | SMTP         | Simple Mail Transfer Protocol |

| No. | Abbreviation | Full Form                                                |
|-----|--------------|----------------------------------------------------------|
| n   | FTP          | File Transfer Protocol                                   |
| o   | HTTP         | HyperText Transfer Protocol                              |
| p   | TCP/IP       | Transmission Control Protocol / Internet Protocol        |
| q   | VoIP         | Voice over Internet Protocol                             |
| r   | DNS          | Domain Name System                                       |
| s   | DSL          | Digital Subscriber Line                                  |
| t   | ADSL         | Asymmetric Digital Subscriber Line                       |
| u   | IPX/SPX      | Internetwork Packet Exchange / Sequenced Packet Exchange |
| v   | ARPANET      | Advanced Research Projects Agency Network                |
| w   | POP          | Post Office Protocol                                     |
| x   | MAC          | Media Access Control                                     |
| y   | ICT          | Information and Communication Technology                 |
| z   | Email        | Electronic Mail                                          |
| aa  | E-fax        | Electronic Fax                                           |
| ab  | ATM          | Asynchronous Transfer Mode                               |

## 5. Multiple Choice Questions

### 1. Telecommunications mainly deals with:

- a. Storing data in computers
- b. Repairing hardware
- c. Sending and receiving data over long distances
- d. Designing software

### 2. A computer network mainly helps computers to:

- a. Increase screen size
- b. Improve graphics
- c. Connect and share resources

d. Reduce electricity use

3. Data packets are:

a. Large storage devices

b. Antivirus programs

c. Small pieces of data sent over a network

d. Hardware chips

4. Frequency is measured in:

a. Volt

b. Watt

c. Hertz

d. Byte

5. Broadband refers to:

a. Slow dial-up connection

b. Cable TV only

c. High-speed Internet connection

d. Satellite TV only

6. Bandwidth is:

a. Actual data speed

b. Delay in transmission

c. Maximum data capacity of a network

d. Variation in signal

7. Latency is measured in:

a. Seconds only

b. Bytes

c. Milliseconds

d. Hertz

8. Compared to 3G, 4G provides:

- a. Slower speed
- b. Same speed
- c. **Faster data speed**
- d. No internet access

9. 5G technology offers:

- a. Lower speed than 4G
- b. Only voice calls
- c. **Very high speed and low latency**
- d. Wired connectivity only

10. Coaxial cable has:

- a. Only one conductor
- b. **Two conductors**
- c. No shielding
- d. Plastic wires only

11. Fiber optic cable transmits data using:

- a. Electrical signals
- b. Magnetic waves
- c. Copper wires
- d. **Light signals**

12. Bluetooth is an example of:

- a. Broadcast radio
- b. **Cellular radio**
- c. Infrared
- d. Satellite

13. RFID uses:

- a. Infrared light
- b. Copper wire
- c. Optical fiber
- d. **Radio waves**

14. RJ45 connector is commonly used with:

- a. Fiber optic cable
- b. Coaxial cable
- c. Twisted pair cable
- d. Satellite dish

15. A media converter works at:

- a. Physical layer
- b. Network layer
- c. Application layer
- d. Transport layer

16. A switch mainly uses:

- a. IP address
- b. MAC address
- c. DNS
- d. URL

17. In bus topology, a terminator is used to:

- a. Increase speed
- b. Boost signal
- c. Stop signal reflection
- d. Encrypt data

18. Ring topology uses:

- a. Token passing
- b. Broadcasting
- c. Random access
- d. Cloud storage

19. In a star topology, all nodes connect to:

- a. Each other directly
- b. A central device
- c. Backbone cable only
- d. Token server

20. PAN covers:

- a. Entire country
- b. Whole city
- c. Short personal range
- d. Multiple continents

21. MAN covers:

- a. Personal room
- b. Single building
- c. Entire globe
- d. City area

22. In P2P architecture:

- a. A central server is required
- b. All computers are equal
- c. Only one computer shares
- d. Only LAN allowed

23. Which of the following is a valid example of an IPv4 address?

- a. 2001:0db8:85a3::8a2e:0370:7334
- b. 192.168.1.1
- c. FE80::0202:B3FF:FE1E:8329
- d. 1200:0000:AB00:1234

24. Which of the following is a valid example of an IPv6 address?

- a. 172.16.0.1
- b. 10.0.0.256
- c. 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- d. 255.255.255.0

25. Ping command is used to:

- a. Check connectivity
- b. Format disk
- c. Install OS
- d. Encrypt data

26. An intranet is:

- a. Public global network
- b. Open social media
- c. Private organizational network
- d. Satellite system

27. An intranet is mainly used within:

- a. A single organization
- b. The entire world
- c. Satellite systems
- d. Mobile towers

28. A communication protocol defines:

- a. Hardware price
- b. Cable color
- c. Screen resolution
- d. Rules for data exchange

29. Which of the following best describes an extranet?

- a. A public social media site
- b. A private network accessible only inside one office
- c. A private network that gives limited access to outsiders
- d. A wireless personal network